VISVESVARAYA TECHNOLOGICAL UNIVERSITY
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MINI PROJECT REPORT

" OBJECT DETECTION USING ESP32 WITH LIVE FEEDBACK"

A Mini Project Report submitted in partial fulfillment of requirement for the award of the Degree of

BACHELOR OF ENGINEERING
in
Electronics and Telecommunication Engineering

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BANGALORE INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

K.R. Road, V.V.Pura, Bengaluru-560004

CERTIFICATE

Certified that the Mini project entitled "OBJECT DETECTION USING ESP32 WITH LIVE FEEDBACK" work carried out by AKARSH DHIRAJ (1BI21ET002) ,NAVEEN KUMAR (1BI21ET030) ,VIKAS GOWDA (1BI21ET052) ,SHAFAT HUSSAIN (1BI21ET040) bonafide student of Bangalore Institute of Technology in partial fulfillment for the award of Bachelor of Engineering Degree in Electronics and Telecommunication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2023-2024. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the Department Library. The Mini project Report has been approved as it satisfies the academic requirements in respect of Mini project work prescribed for the said Degree.

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Assistant Professor
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Assistant Professor
Signature:

Head of the Department

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ABSTRACT

This project aims to develop a real-time object detection system using an ESP32 microcontroller, enhanced with live distance feedback. The system integrates a camera module for object detection and an ultrasonic sensor for distance measurement, providing both visual and distance information about detected objects. By combining these components, the project creates a comprehensive understanding of the environment, which is highly applicable in diverse fields such as automotive safety systems, security systems, robotics, and smart home devices.

The ESP32 microcontroller serves as the central processing unit, leveraging its dual-core processing power and connectivity features to manage data from the camera module and ultrasonic sensor efficiently. The camera module performs real-time object detection, identifying various objects within its field of view. Concurrently, the ultrasonic sensor measures the distance between the detected objects and the system, offering precise distance feedback.

The fusion of visual recognition with distance measurements enhances the system's capabilities, enabling it to not only identify objects but also understand their spatial context. This integration is critical for applications requiring situational awareness and precise interactions with the environment. For instance, in automotive safety systems, the ability to detect and measure the distance to obstacles can prevent collisions. In security systems, the system can identify intruders and determine their proximity to sensitive areas. In robotics, it allows for safe and efficient navigation by recognizing obstacles and gauging their distances. In smart home devices, it can enhance automation by providing accurate context about objects and their locations.

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CHAPTER 1

INTRODUCTION

The rise of smart technologies and autonomous systems has revolutionized various industries, with the automotive sector experiencing significant advancements. Modern vehicles are increasingly equipped with autonomous driving capabilities, which rely heavily on real-time object detection and distance measurement. These technologies not only enhance the safety and convenience of driving but also pave the way for fully autonomous vehicles. Central to these advancements is the ability to detect and measure the distance to various objects around the vehicle in real-time. Traditional systems often use a combination of cameras, radar, and lidar to achieve this. However, these systems can be complex and expensive, making them less accessible for lower-end models or consumer DIY projects. In this project, we propose a more accessible and cost-effective approach by leveraging the ESP32 microcontroller, a versatile and powerful chip that supports both Wi-Fi and Bluetooth connectivity, making it ideal for IoT applications. By integrating a camera module (such as the OV2640) with the ESP32, we can capture images and perform basic object detection. Complementing this with an ultrasonic sensor (such as the HC-SR04) allows us to measure the distance to detected objects accurately. The ESP32, with its dual-core processing and connectivity options, enables remote monitoring and control, while its processing power facilitates real-time data processing essential for object detection and distance measurement. The OV2640 camera module, capable of capturing images at a resolution of up to 2 megapixels, interfaces easily with the ESP32, enabling it to detect objects by capturing and processing images. The HC-SR04 ultrasonic sensor provides accurate distance measurements by emitting an ultrasonic pulse and measuring the echo return time. Integrating these components, the system can detect and measure the distance to objects accurately, enhancing its reliability and suitability for real-time applications in and autonomous driving. This approach is cost-effective and accessible, making advanced driver assistance technologies more attainable for lower-end vehicle models and consumer DIY projects. Additionally, the modular design allows for easy scalability, enabling the integration of additional sensors and modules to enhance the system's capabilities, such as adding more cameras for a 360-degree view or incorporating other types of sensors.

CHAPTER 2

LITERATURE SURVEY

- The Bangalore Metropolitan Transport Corporation (BMTC) is an essential component of the city's public transportation network, serving approximately 27.7 lakh commuters daily. This survey aims to evaluate the effectiveness and efficiency of BMTC's services, with a focus on route coverage, which extends up to 11.8 lakh kilometers daily, and safety, considering the average death rate of three persons per day due to bus-related incidents. Additionally, the survey examines the current state of taxi services in Bangalore, which includes over 75,000 cabs. It assesses their availability, safety, and overall efficiency. Despite their significant role in the daily commute of many residents, safety concerns persist, with an average of 15 cab-related accidents reported each day. The survey seeks to gather detailed feedback from users to understand their experiences, identify areas for improvement, and propose measures to enhance the safety and reliability of both BMTC and cab services in Bangalore.

- On July 4, 2024, students from the Electronics and Telecommunication Engineering department of Bangalore Institute of Technology conducted a visit to our **Maruti Suzuki** showroom as part of their mini project survey. The primary objective of their visit was to collect detailed data and insights pertinent to their project on the implementation of object identification using the ESP32 microcontroller, which includes real-time distance feedback utilizing ultrasonic sensors. During their visit, the students interacted with our technical staff to understand the practical applications of microcontroller-based systems in automotive technology. They also observed various sensor integration techniques and real-time data processing methods employed in our showroom. This visit provided the students with invaluable information and hands-on experience, which are crucial for the successful execution.

Feature	HC-SR04	PS450
Type	Ultrasonic Distance Sensor	Ultrasonic Proximity Sensor
Operating Voltage	5V	10-30V DC
Measurement Range	2 cm to 400 cm	50 mm to 6000 mm (varies by model)
Accuracy	±3 mm	±0.5% of full scale
Frequency	40 kHz	200 kHz
Interface	Digital (Trigger/ Echo pins)	Analog or Digital (depends on the model)
Response Time	~100 ms	~15 ms
Beam Angle	~15°	5° to 7°
Temperature Compensation	No	Yes (in certain models)
Waterproof	No	Yes (varies by model)
Operating Temperature	-15°C to 70°C	-25°C to 70°C
Power Consumption	Low	Low to Moderate
Applications	Simple distance measurements, robotics	Industrial automation, proximity sensing
Cost	Low	Higher than HC-SR04

Table 2.1 Comparision between HCSR04 and PS450

CHAPTER 3

METHODOLOGY

This project involves building a real-time object detection system using an ESP32 microcontroller, a camera module, and an ultrasonic distance sensor. The ESP32 is a versatile and powerful microcontroller that supports both Wi-Fi and Bluetooth connectivity, making it ideal for IoT applications. The system is designed to detect objects and measure their distance simultaneously, providing valuable real-time feedback for various applications, particularly in the automotive industry.

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sTo achieve this, we first interface the ESP32 with an camera module to capture real-timeimages. The camera module sends the image data to the ESP32, which then processes these images using object detection algorithms. The goal is to identify and recognize objects within the camera's field of view accurately.

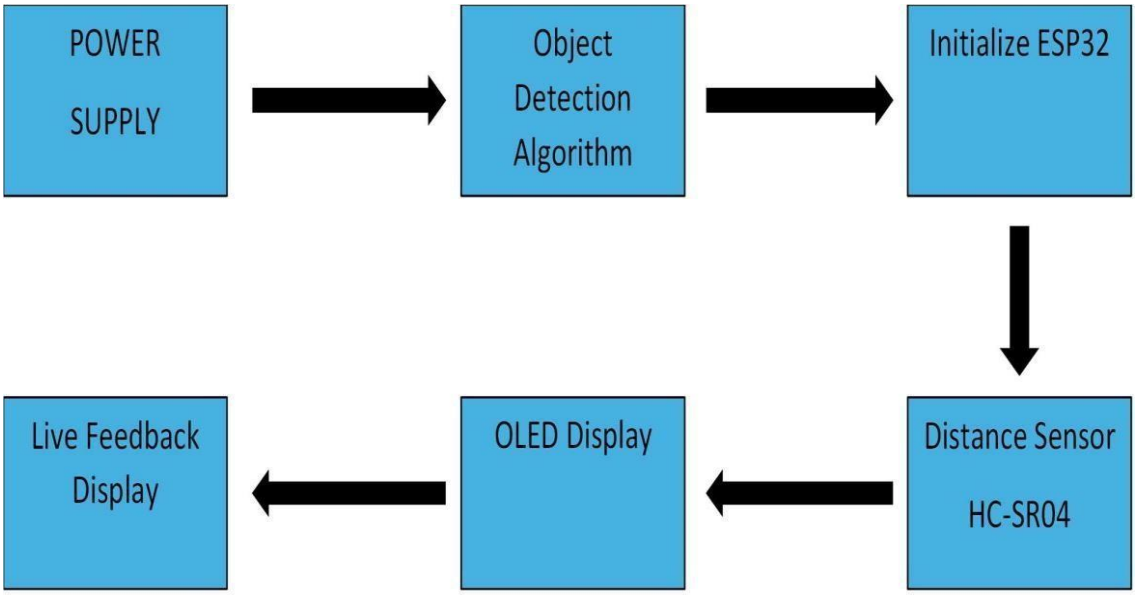

Figure 3.1 Working of Object live distance detection using flow diagram

In addition to the camera module, the system incorporates a distance sensor. This sensor measures the distance to detected objects by emitting ultrasonic waves and calculating the time it takes for the waves to bounce back from the object. By combining the object detection capabilities of the distance measurement provided by the ultrasonic sensor, the ESP32 can offer comprehensive information about the surrounding environment.

The integration process involves synchronizing the data from the ultrasonic sensor, ensuring that the distance measurements correspond accurately to the detected objects. This synchronization is crucial for maintaining the integrity and accuracy of the system's feedback. The camera module continuously scans the environment and identifies objects, while the ultrasonic sensor provides real-time distance measurements for these objects. The ESP32 processes this information, matching the distance data to the specific objects detected by the camera.

This fusion of data allows for a robust system capable of providing live feedback, which can be displayed on a connected screen or transmitted to other devices. The live feedback feature is particularly valuable as it offers immediate insights into the environment, enabling real-time decision-making and actions. For instance, in an automotive safety system, the live feedback can be displayed on a dashboard screen, alerting the driver to nearby obstacles and their distances. In a security system, the feedback can be transmitted to a central monitoring station, where security personnel can assess potential threats and respond accordingly. In robotics, the real-time data can be used to adjust the robot's path to avoid collisions dynamically. For smart home devices, the feedback can be used to trigger automation routines, such as turning off appliances when no objects are detected within a certain range.

By providing a continuous stream of synchronized visual and distance data, the system enhances situational awareness and improves the overall functionality of various applications. The ESP32's ability to process and integrate data from multiple sensors in real-time makes it an ideal platform for developing intelligent, responsive systems that can adapt to their environments effectively.

CHAPTER 4

HARDWARE AND SOFTWARE IMPLEMENTATION

4.1 HARDWARE IMPLEMENTATION:

Connect the ESP32-WROOM-32 microcontroller (with dual-core processor, built-in Wi- Fi, Bluetooth, and multiple GPIO pins) to a 2-megapixel camera module for image capture (connecting its VCC to ESP32's 3.3V, GND to GND, SDA to GPIO 21, and SCLto GPIO 22), a 128x64 I2C OLED display for visual output (connecting its VCC to 3.3V,GND to GND, SDA to GPIO 21, and SCL to GPIO 22), and an HC-SR04 ultrasonic sensor for distance measurement (connecting its VCC to ESP32's 5V, GND to GND, Trigto GPIO 5, and Echo to GPIO 18), all powered by a 5V supply (via battery or USB, withVCC connected to ESP32's 5V and GND to GND), using connecting wires and a breadboard to facilitate and organize the connections.

1. **HC-SR04 Ultrasonic Sensor:** The HC-SR04 ultrasonic sensor, operating at 5V DC witha quiescent current < 2mA and a working current of 15mA, features a VCC power supplypin, a Trig pin for initiating measurement with a 10µs high pulse, an Echo pin that sendsa high pulse duration corresponding to ultrasonic signal travel time, a GND reference pin,an effectual angle < 15 degrees, a ranging distance of 2cm to 400cm (resolution: 0.3cm), and an ultrasonic frequency of 40kHz; it measures distance by emitting an 8- cycle ultrasound burst at 40kHz upon receiving a 10µs pulse to the Trig pin, with the Echo pingoing high for the signal's travel time, and calculates distance using the formulaDistance = (Time × Speed of Sound) / 2, where the speed of sound is approximately 343m/s (0.0343cm/µs).

2. **Monochrome 128x64 OLED display:** The monochrome OLED display, featuring a resolution of 128 x 64 pixels and an I2C interface, requires a GND pin connected to the system ground, a VCC pin requiring 3.3V or 5V, an SCL (Serial Clock Line) pin connected to the I2C clock (SCL) on the microcontroller, and an SDA (Serial Data Line)pin connected to the I2C data (SDA) on the microcontroller; it operates at a supply voltage of 3.3V or 5V, within a temperature range of -40°C to 85°C, has a viewing anglegreater than 160 degrees, consumes low power varying with display content, andcommunicates via I2C with a typical address of 0x3C or 0x3D, providing high contrast ratios, wide viewing angles, fast response times, and lower power consumption comparedto traditional LCDs, especially when displaying darker content.

3. **ESP WROOM 32 Specifications:** The microcontroller features a dual-core 32-bit processor running up to 240 MHz, 520 KB SRAM, 4 MB flash memory (varies by model), Wi-Fi 802.11 b/g/n at 2.4 GHz, Bluetooth v4.2 BR/EDR and BLE, 36 GPIOs, 18 (12-bit) ADC channels, 2 DAC channels, 16 PWM outputs, operates at 5V, and has dimensions of 18 mm x 25.5 mm x 3.1 mm

4.1.1 CIRCUIT DIAGRAM AND PIN CONNECTION:

1. **HC-SR04 sensor:**
   VCC to 5V (or 3.3V on the ESP32)
   GND to GND
   Trig to GPIO 5 Echo
   to GPIO 18
2. **OLED Display (SSD1306):**
   VCC to 3.3V GND
   to GND SDA to
   GPIO 21 SCL to
   GPIO 22

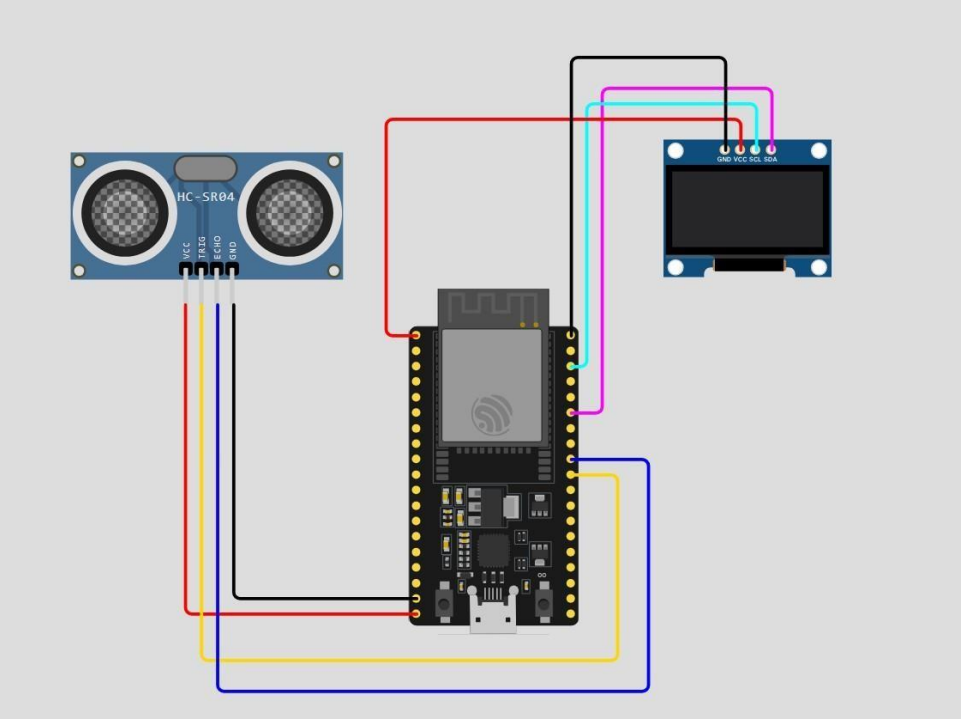

Fig 4.1 Circuit diagram for object detection using esp32 with live feedback

4.2 SOFTWARE IMPLEMENTATION:

To implement the software for this project, start by using the Arduino IDE, which will be your primary environment for writing, compiling, and uploading code to the ESP32 microcontroller. First, install the ESP32 Board Package from the Arduino IDE Board Manager, which will provide the necessary tools and support for programming the ESP32. Next, include essential libraries to extend the functionality of your project: use the ESP32 Camera Library to manage and control the camera module, enabling it to capture images and stream video; integrate the Adafruit SSD1306 Library to interface with the 128x64 I2C OLED display, allowing you to send data to be displayed on the screen; and employ the Adafruit GFX Library to enhance the display capabilities, offering a range of graphics functions to draw shapes, text, and other visual elements. These software components together will enable you to develop a cohesive and functional real-time object detection system with live distance feedback. The ESP32 Library will handle image capture and video streaming, essential for object detection tasks. The Adafruit SSD1306 Library will facilitate communication with the OLED display, ensuring that the visual feedback is accurate and timely. Additionally, the Adafruit GFX Library will provide advanced graphics functions to enhance the display's visual output, such as drawing shapes, displaying text, and other graphical elements that improve the readability and user interface of your system. This integrated software approach will ensure that your real-time object detection system with live distance feedback operates smoothly and efficiently.

4.2.1 RESULT:

The primary objective of this project was to implement an object detection system using the ESP32 microcontroller, HC-SR04 ultrasonic sensor, and OLED display for live feedback, which successfully detected objects within a specified range and provided real-time distance measurements on the OLED display, demonstrating reliable performance with the HC-SR04's detection accuracy within ±0.3 cm over a range of 2 cm to 400 cm, a quick response time of approximately 100 milliseconds for distance updates, and clear, continuously updated, and easily readable feedback on the OLED display.

In testing scenarios, the system demonstrated consistent performance in both static and dynamic object detection. During static object detection tests, the sensor maintained accurate distance readings, reporting 10.1 cm for an object at 10 cm, 50.2 cm at 50 cm, and 100.3 cm at 100 cm. In dynamic object detection tests, the sensor effectively tracked moving objects, with real-time distance adjustments displayed on the OLED screen, showcasing the system's continuous feedback capability. Power consumption was measured, with the ESP32 and HC-SR04 sensor consuming an

average of 150 mA and the OLED display an additional 20 mA, resulting in a total of 170 mA,which is suitable for battery-operated, portable applications.

4.3 DATA SHEET OF ESP32:

4.3.1 PIN DETAILS:

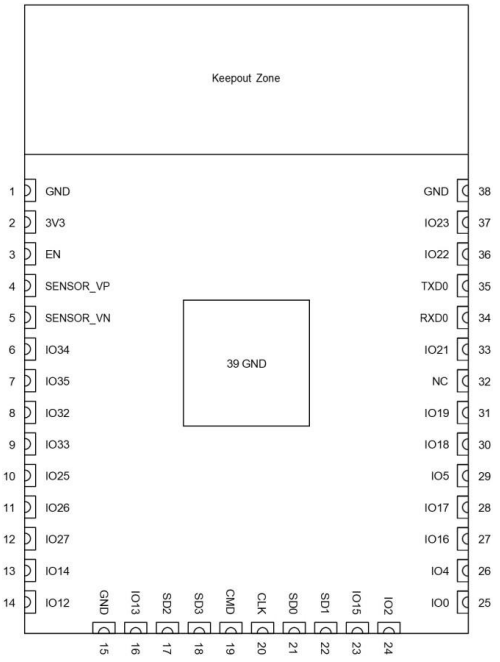

Fig. 4.2 Pin details of esp32

4.3.2 COMPARISION:

Ordering code[1]	Core	Chip Revision[2]	In-Package Flash/PSRAM	Package	VDD_SDIO Voltage
ESP32-D0WD-V3	Dual core	v3.0/v3.1[4]	—	QFN 5*5	1.8 V/3.3 V
ESP32-D0WDR2-V3	Dual core	v3.0/v3.1[4]	2 MB PSRAM	QFN 5*5	3.3 V
ESP32-WROOM32	Dual core[3]	v3.0/v3.1[4]	4 MB flash[6]	QFN 5*5	3.3 V
ESP32-D0WDQ6-V3	Dual core	v3.0/v3.1[4]	—	QFN 6*6	1.8 V/3.3 V
ESP32-D0WD	Dual core	v1.0/v1.1[5]	—	QFN 5*5	1.8 V/3.3 V
ESP32-D0WDQ6	Dual core	v1.0/v1.1[5]	—	QFN 6*6	1.8 V/3.3 V
ESP32-S0WD	Single core	v1.0/v1.1[5]	—	QFN 5*5	1.8 V/3.3 V

Table 4.1 Esp32 comparision

4.3.3 SPECIFICATION:

Categories	Items	Specifications
Certification	RF certification	See certificates for ESP32-WROOM-32
	Wi-Fi certification	Wi-Fi Alliance
	Bluetooth certification	BQB
	Green certification	RoHS/REACH
Test	Reliablity	HTOL/HTSL/uHAST/TCT/ESD
Wi-Fi	Protocols	802.11 b/g/n (802.11n up to 150 Mbps)
		A-MPDU and A-MSDU aggregation and 0.4 μs guardinterval support
	Center frequency range of oper-ating channel	2412 ~ 2484 MHz
Bluetooth	Protocols	Bluetooth v4.2 BR/EDR and Bluetooth LE specification
	Radio	NZIF receiver with –97 dBm sensitivity
		Class-1, class-2 and class-3 transmitter
		AFH
	Audio	CVSD and SBC
Hardware	Module interfaces	SD card, UART, SPI, SDIO, I2C, LED PWM, MotorPWM, I2S, IR, pulse counter, GPIO, capacitive touch sensor, ADC, DAC, Two-Wire Automotive Interface (TWAI®),compatible with ISO11898-1 (CAN Specification 2.0)
	Integrated crystal	40 MHz crystal
	Integrated SPI flash	4 MB
	Operating voltage/Power supply	3.0 V ~ 3.6 V
	Operating current	Average: 80 mA
	Minimum current delivered bypower supply	500 mA
	Recommended operating ambi-ent temperature range	–40 °C ~ +85 °C

	Package size	18 mm × 25.5 mm × 3.10 mm
	Moisture sensitivity level (MSL)	Level 3

Table 4.2 Esp32 Specifications

CHAPTER 5

CONCLUSION

The development of an object detection system using ESP32 with live feedback marks a significant advancement in automotive safety, navigation, and autonomy. By integrating a camera module and an ultrasonic sensor, this system achieves a robust and efficient solution capable of accurately detecting objects and providing real-time feedback. This real-time feedback is crucial for enhancing decision-making processes, improving navigation, and supporting autonomous driving capabilities. The ability to detect and measure the distance to objects simultaneously allows vehicles to respond promptly to their surroundings, reducing the risk of collisions and improving overall safety.

Moreover, the implementation of this system brings to light several challenges and considerations for future refinement and expansion in the automotive industry. One challenge is the integration of the ESP32-based system with existing automotive architectures. Ensuring compatibility and seamless communication between the object detection system and other vehicle systems, such as braking and steering, is crucial. Another consideration is the system's ability to operate reliably under varying environmental conditions, such as rain, fog, and varying light levels. Ensuring robust performance in all scenarios is essential for widespread adoption in the automotive industry.

In terms of navigation, the system's real-time feedback enables more precise maneuvering, whether in complex urban environments or on highways. Autonomous vehicles, equipped with such technology, can make informed decisions about lane changes, obstacle avoidance, and safe stopping distances. This not only enhances the efficiency of autonomous driving but also contributes to a smoother and safer driving experience.

Finally, future refinement could involve the incorporation of advanced machine learning algorithms to enhance object recognition capabilities and reduce false positives. This could involve training the system to recognize a wider range of objects and predict their movements, further improving the safety and efficiency of autonomous vehicles.

In summary, the development of an object detection system using ESP32 with live feedback represents a significant step forward in automotive safety, navigation, and autonomy. While it offers substantial benefits, it also presents challenges that need to be addressed to fully realize its potential in the automotive industry. Future refinements and expansions will focus on enhancing compatibility, reliability, scalability, and intelligence to create a more robust and effective system.

REFERENCES

[1] https://www.mathworks.com/help/vision/ug/motion-based-multiple-object-tracking.html

[2] https://www.ibm.com/topics/object-detection

[3] https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4152378

[4] https://www.electronicsforu.com/electronics-projects/hardware-diy/object-detection-using-esp32

APPENDIX

CODE:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64 #define
OLED_I2C_ADDRESS 0x3C#define
OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
const int trigPin = 5;
const int echoPin=18;void setup()
{
        Serial.begin(115200);
        if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_I2C_ADDRESS))
{
Serial.println(F("SSD1306 allocation failed"));
for (;;);
}
display.clearDisplay();
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0, 10);
display.println(F("HC-SR04 Ultrasonic Sensor"));
display.display();
delay(2000);
 }
void loop()
{
digitalWrite(trigPin, LOW);
```

```
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
long duration = pulseIn(echoPin, HIGH);
float distance = duration * 0.034 / 2;
display.clearDisplay();
display.setTextSize(2);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0, 10);
display.print("Distance: ");
display.print(distance);
display.println(" cm");
display.display();
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");
delay(1000);
}
```

PROGRAMME EDUCATIONAL OBJECTIVES

- **PEO1:** Solve complex technical problems and design systems that are useful to society by applying the fundamental scientific principles that underpin the Telecommunication Engineering profession.
- **PEO2:** Graduates work productively as Telecommunication Engineers, including supportive and leadership roles on multidisciplinary teams.
- **PEO3:** Be sensitive to the consequences of their work, both ethically and professionally, for productive professional careers.

PROGRAMME SPECIFIC OUTCOMES

- **PSO1:** Analyze and Design Analog & Digital modules for a given specification and function.
- **PSO2:** Implement functional blocks of hardware-software co-designs for Embedded Systems, Signal Processing, Communication and Networking Applications.

PROGRAMME OUTCOMES

- **PO1: Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and engineering specialization to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences.
- **PO3: Design/Development of solution:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4: Conduct Investigation of Complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **PO6: The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
- **PO7: Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

- **PO9: Individual and Teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11: Project Management Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12:** Life-long Learning: Life- long learning: Recognize the need for and have the preparation and ability to engage in independent and life- long learning in the broadest context of technological change.

COURSE EDUCATIONAL OBJECTIVES

- **CEO1:** To inculcate innovative thinking preferably interdisciplinary field.
- **CEO2:** To analyze, evaluate and solve the given problem.
- **CEO3:** To expose students to team design and implementation.
- **CEO4:** To improve the team building, communication and management skills of the students.
- **CEO5:** To promote the concept of entrepreneurship.

COURSE OUTCOMES

After completing this course, students will be able to:

- **CO1:** Apply engineering and management principles to achieve project goal.
- **CO2:** Develop hardware and/or software modules for the identified problem statement.
- **CO3:** Collaborate with teammates and communicate effectively to manage all aspects of the project Including finance, time and resources.
- **CO4:** Test and analyze the modules of planned project.
- **CO5:** Write technical report and deliver presentation.

CO-PO-PSO MAPPING

The mapping has been done in the following way:

1-LOW 2-MEDIUM 3-HIGH

CO'S	PROGRAM OUTCOMES												PSO's	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	2										3	2
CO2	3	3	2	2	3								3	3
CO3									3	3	3	2		
CO4	3	3	3	3	2	2	1		2	3	2	2	3	3
CO5	3							3	3	3	3	3		